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The Fundamental Universe

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The universe is vast, chaotic and it contains an incredible number of stars and planets – all of which are formed from atomic elements that can be further divided into subatomic particles. Although the universe appears to be a complex mix of particles, a simpler definition is possible when the components of the universe are split into fundamental parts. This definition, and how it applies to matter, forces and time is explored using a concept of wave energy and fundamental waves that travel the universe.

1. INTRODUCTION

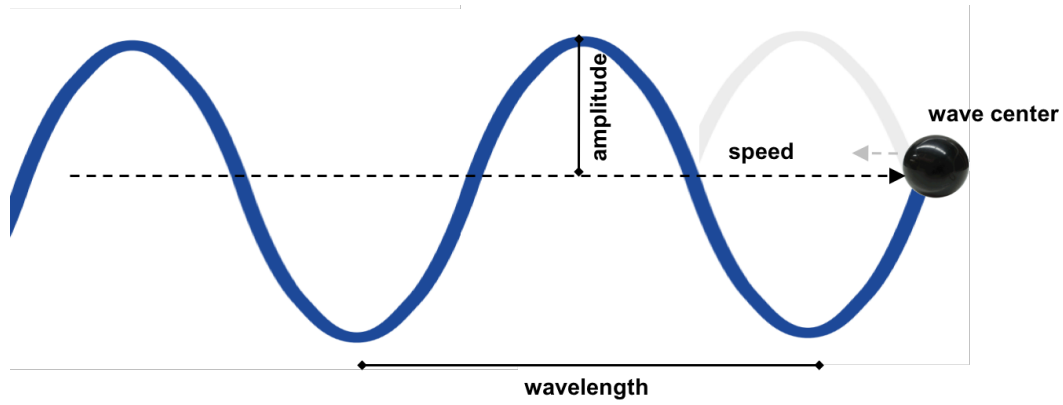
Although the universe contains billions of galaxies, and each of these galaxies contains billions of stars, the matter that makes up these stars and planets in the galaxies is consistent. Matter is formed from the 118 known atomic elements of the Periodic Table of Elements [1]. Of these atomic elements, there is a fundamental nucleon that is responsible for the creation and unique properties of each of these elements – the proton. The word *fundamental*, in this case, is used to describe the foundation of atomic elements. When paired with an electron, one single proton forms the atomic element known as hydrogen. Hydrogen is the simplest element. Yet it becomes the foundation of all elements when more protons are added to the atomic nucleus. The proton is the fundamental unit of elements.

This same logic can be applied to other properties of the universe. The universe contains fundamental units that provide the foundation of particles, charges and even time itself. In this definition, fundamental is a default value of a certain property in the universe, but one in which the property may change.

The properties of the universe that are discussed in this paper are based on foundational components of waves. Particles and photons are known to have wave properties, so it is logical to consider the impact of the components of waves and how speed, wavelength and amplitude changes affect matter and light. Each of these are described as fundamental – with a default value – but each is considered as a foundation and is subject to change under the right conditions.

2. THE FUNDAMENTALS

The fundamental universe consists of waves with default values for wave speed, wavelength and wave amplitude. It will be shown in sections 2.1 to 2.3 how these properties change and become the foundation for matter, light and their interactions. Section 2.4 introduces a property called the wave center, introduced by Dr. Milo Wolff, which reflects a wave to become a standing wave [2]. First, an illustration of these wave properties is provided in Figure 1.



longitudinal wave represented as a sinusoidal wave

Fig. 1. Properties of a longitudinal wave: wavelength, amplitude and speed, reflected by a wave center.

2.1. Wave Speed & The Medium

Waves have a known speed in the universe, otherwise known as the speed of light. Waves in the universe travel at a speed of 299,792,458 meters per second. In physics, this speed is a constant – a fixed value. In this paper, this speed is considered a fundamental property.

In empty space, light travels at the speed of light (c). However, in a medium such as water or air, light slows down. A medium with molecules affects a photon's speed, and its speed depends on the density, or the number of molecules, in a given space. Therefore, wave speed must be considered fundamental and is subject to change based on the density of the medium in which the wave travels. A wave can only travel at the speed of light when the medium is empty space.

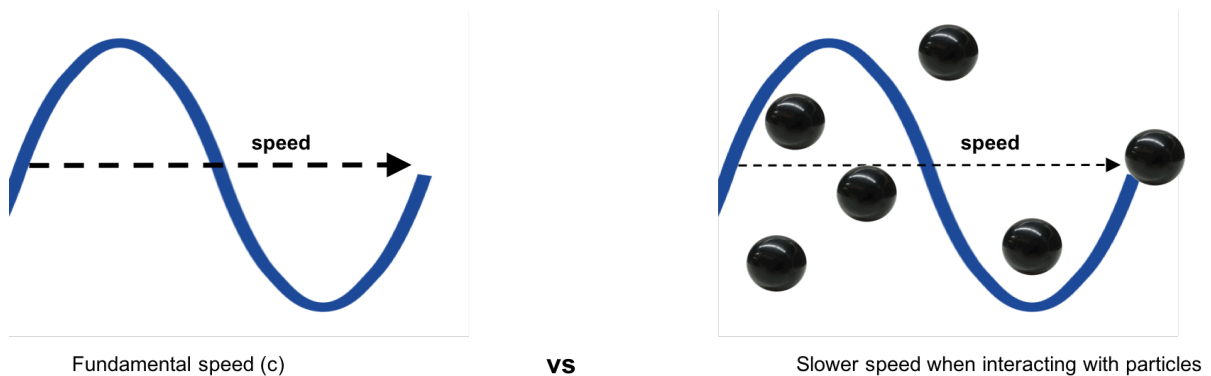


Fig. 2. Fundamental speed vs a slower wave speed in a denser medium.

2.2. Wavelength & Time

In the *Particles of the Universe 2*, the universe is described as traveling longitudinal wavelets that form wavefronts according to Huygen's principle [3]. If this is indeed the case, there is a fundamental wavelength for longitudinal waves. The particles that form matter, such as the electron, are known to have wave properties. This is referred to as wave-particle duality. In the *Wave Structure of Matter (WSM)*, Wolff describes and calculates the electron as a standing wave of energy [2]. This definition of particles, as standing longitudinal waves of energy, provides a new definition of time due to a fundamental wavelength for these *longitudinal* waves. For clarity, the photon, which is responsible for the electromagnetic wave, is a *transverse* wave with a variable wavelength as a result of particle vibration.

A fundamental wavelength for the longitudinal waves that travel the universe provides the mechanism for time – a universal clock. If these waves have a fundamental wave speed (c) and wavelength (λ), then there is a fundamental frequency (f) as described mathematically:

$$f = \frac{c}{\lambda} \quad (1)$$

Frequency is a measurement of time, literally being the inverse of a time unit. These longitudinal waves flow through all of matter in space with a fundamental wavelength and frequency – creating a clock by which everything is run. These waves flow through the matter that makes up DNA and the biological clocks of living things. They flow through subatomic particles that cause the repetition of decay for the most accurate atomic clocks. They flow through the matter of large celestial bodies that cause the rotation of the Earth and its orbit around the Sun causing days and years. There is a consistent, universal clock provided by these traveling longitudinal waves in space. The fundamental wavelength is responsible for time.

Yet we know that time is relative to an observer. Wavelength is another property that has a fundamental value that can change under certain conditions – in this case it is the motion of matter. When an object is in motion, its wavelength changes relative to an observer. This is a property of waves known as the Doppler effect [4]. The geometric mean wavelength of particles, which are standing waves of energy, becomes shorter when an object is in motion. Thus, time slows on this object relative to a stationary object.

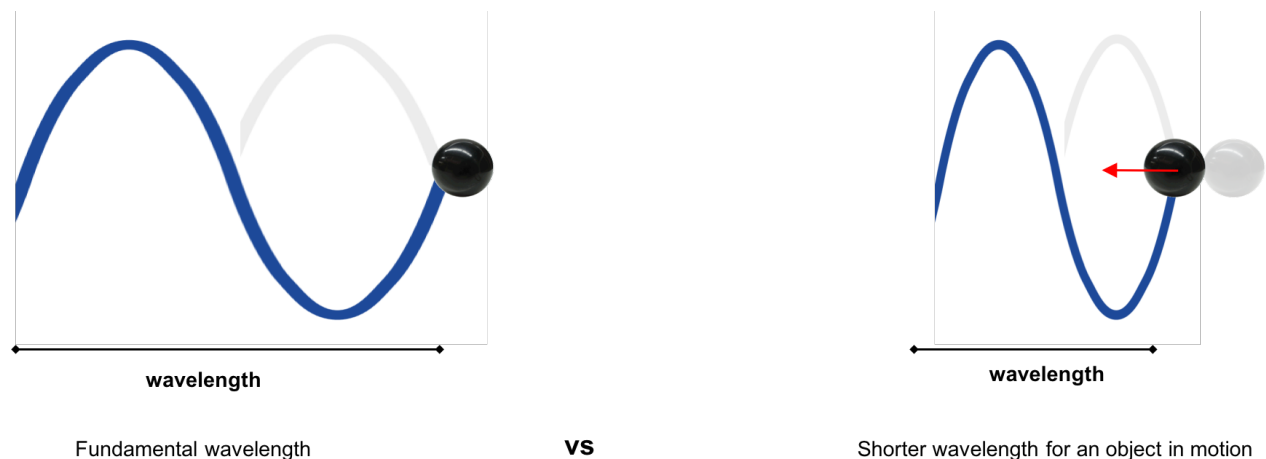


Fig. 3. Fundamental wavelength vs a shorter geometric mean wavelength for an object in motion (relativity).

2.3. Wave Amplitude & Charge

A fundamental longitudinal wave would also have a fundamental wave amplitude. The three properties to describe a wave all have universal (default) values: speed, wavelength and amplitude. In Wolff's work on the electron, the movement of particles is based on a law that they move to minimize wave amplitude. Amplitude is responsible for a particle's charge – it is the electric force that causes a movement of a particle to minimize this wave amplitude.

Waves are known to experience constructive and destructive interference, which affects amplitude [5]. Two waves in the same phase are constructive, adding their wave amplitudes. Two waves that are phase shifted by 180 degrees are destructive, subtracting wave amplitude. These are the possible nodes on the wavelength of a standing wave, where nodes are the point of minimal wave amplitude. Particles that are in the same phase have the same charge; particles that are phase shifted are the anti-particles

with opposite charge. When particles have the same charge, wave amplitude increases between particles due to constructive wave interference, forcing the particles away from each other. For example, two electrons combine their wave amplitudes between particles and repel. When particles have opposite charges, wave amplitude decreases between the particles due to destructive wave interference, attracting the particles. For example, an electron and positron have destructive waves and are attractive because of the rule that a particle moves in the direction of minimal wave amplitude.

Wave amplitude is fundamental. There is a default wave amplitude for longitudinal waves, yet a combination of particles creates a new wave amplitude due to constructive wave interference. Once again, fundamental is a default value but subject to change based on a condition. Two electrons in proximity will have twice the wave amplitude due to constructive wave interference (illustrated in Figure 4). Ten electrons in proximity will have ten times the wave amplitude for the same reason.

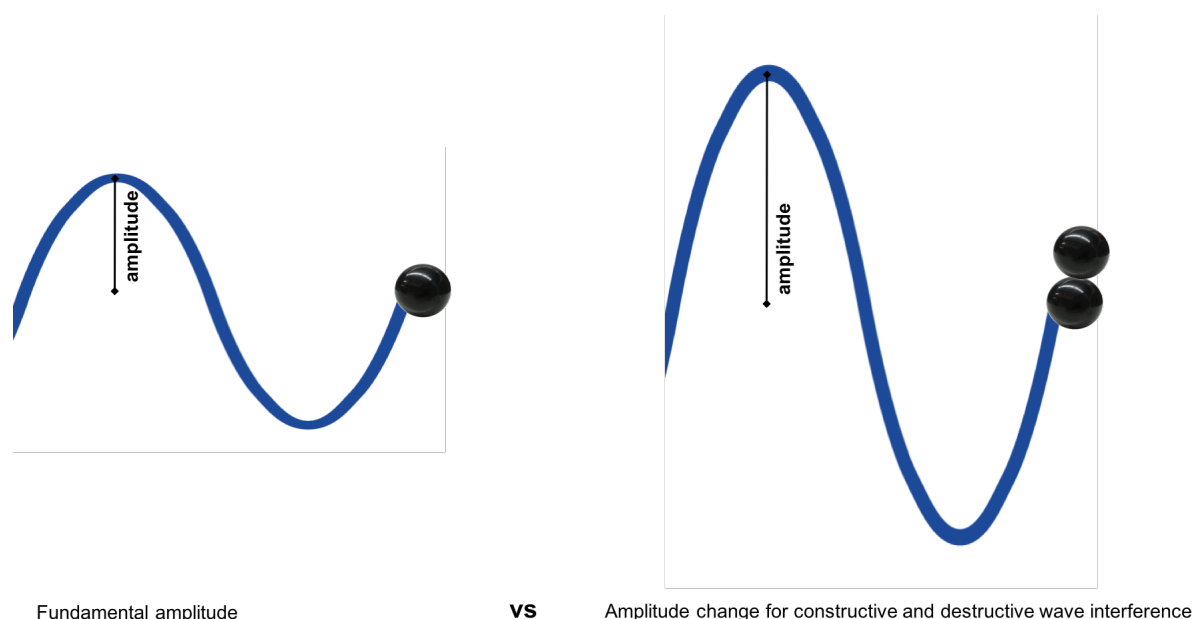


Fig. 4. Fundamental amplitude vs change in amplitude from wave interference.

2.4. Wave Centers & Particles

Wolff's work on the standing wave structure of particles includes the concept of a wave center. The wave center is a point at which incoming waves (in-waves) are reflected to become out-waves. The combination of in-waves and out-waves creates a standing wave until the particle's radius, at which they break down to become traveling waves once again. This structure is three-dimensional, with wavelets that form a spherical wavefront according to Huygen's principle. The result is a spherical, standing wave of stored energy as a result of a wave center that reflects longitudinal waves. This becomes the definition of mass – it is stored energy as a result of the natural property of standing waves. Energy travels as waves; the mass of matter is a specific form of waves known as standing waves.

Although Wolff's explanation of the wave center was focused on the electron particle, a single wave center is more likely to be the lightest-known subatomic particle, called the neutrino. In *The Relation of Particle Sequence to Atomic Sequence*, a fundamental particle was calculated to have a rest energy value of 2.39 electron-volts (eV) [6]. This is near the upper-end of a predicted neutrino range of 2.2 eV [7]. In this work, a fundamental particle was calculated by taking the known rest energies of subatomic particles, ranging from the neutrino to the Higgs boson, and by creating a linear function of their energies against a particle number (K). The particle number is a variable based on a collection of wave centers, combining to create a new particle. Figure 5 illustrates a combination of wave centers and its effect on amplitude and particle radius to cause a change in stored (standing wave) energy.

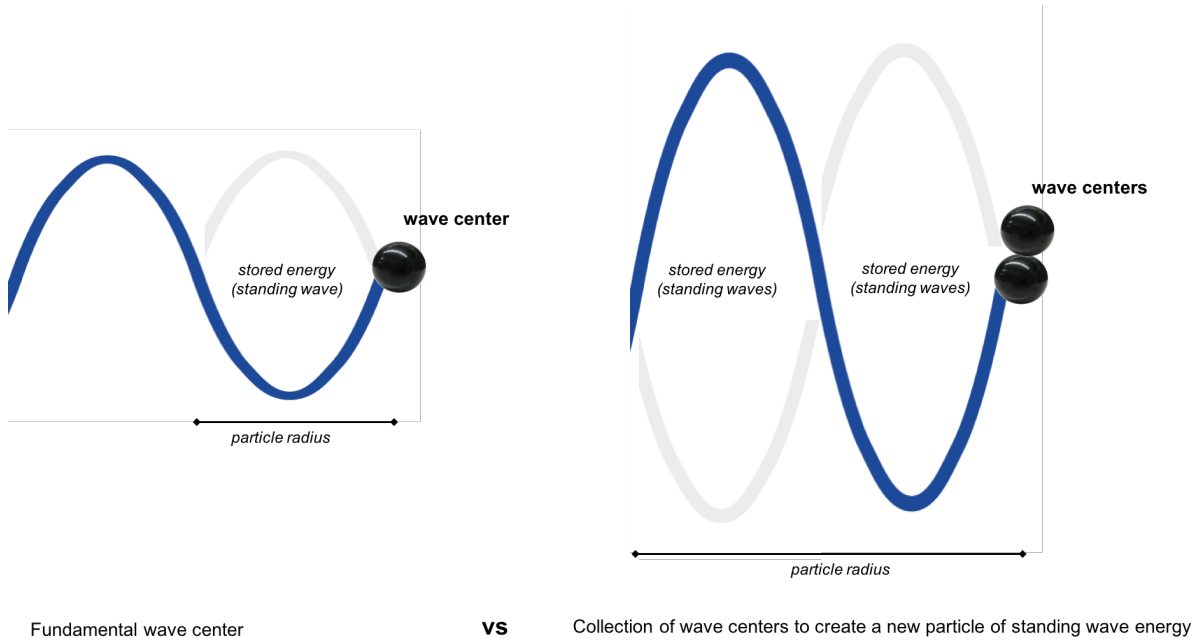


Fig. 5. Fundamental wave center vs a collection of wave centers to create new particles.

The linear relation of a particle number to particle energy is identical to the process of mapping atomic number to atomic mass. The latter is the linear function of the mass of elements when protons are added to a nucleus. Unlike atomic elements, however, the linear function for particles is based on the fifth power of the particle number (K), as represented by the following equation where E is energy, measured in electron-volts [6]:

$$E = (2.39) K^5 \sum_{n=1}^K \frac{n^3 - (n-1)^3}{n^4} \quad (2)$$

The previous equation for particles is represented graphically in Figure 6 and compared to the linear equation for atomic elements. The detailed graphs are found in *The Relation of Particle Sequence to Atomic Sequence*.

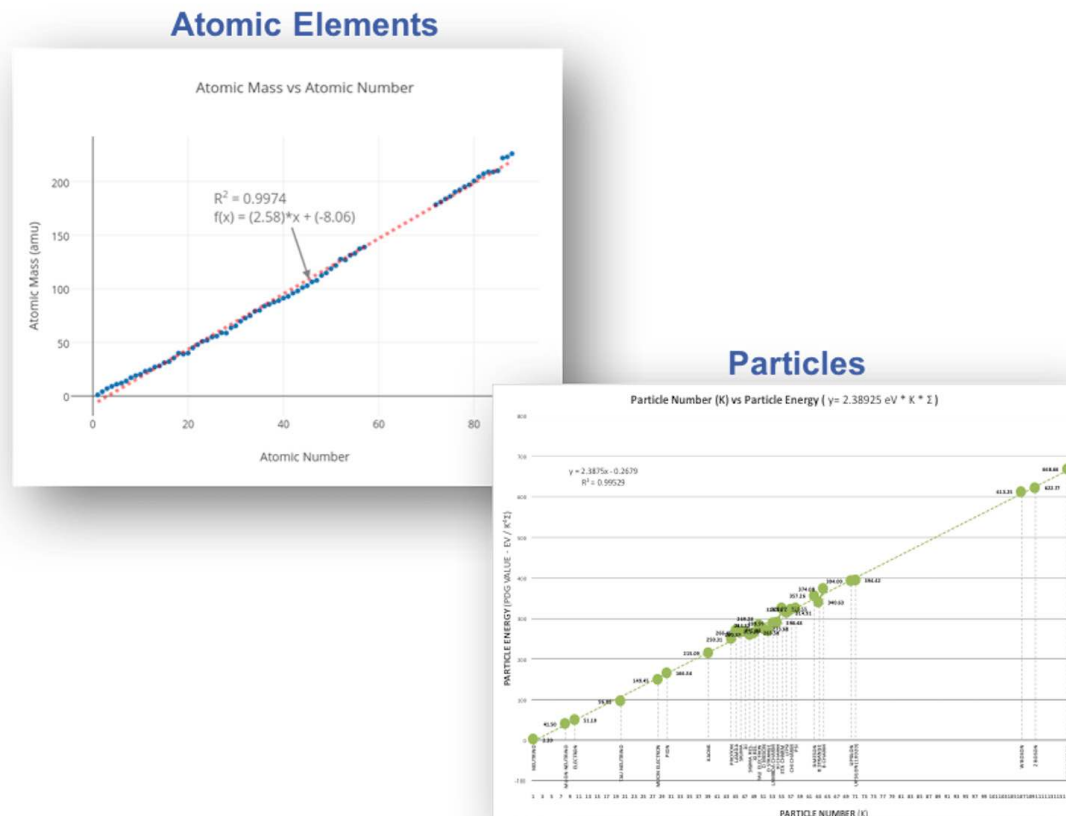


Fig. 6. Atomic Mass vs Atomic Number (linear function) & Particle Energy vs Particle Number (linear function) [6].

Fundamental, in this case, is a single wave center that reflects longitudinal waves to create the standing wave structure of a single particle, quite possibly the neutrino. It is the fundamental particle. Under certain conditions – when fundamental particles combine – new particles are created from standing wave energy due to a change in wave amplitude and standing wave distance (particle radius). The fundamental particle is the foundation to create other particles, just like a single proton is the foundation to create atomic elements.

3. CONCLUSION

The word *fundamental* has its roots in the Latin word fundamentum, which means foundation [8]. In the context of this paper, fundamental is the foundation for energy, the creation of particles and matter, the motion and forces of particles and the definition of time. It begins with a belief that energy travels in the form of waves in the universe, and these waves have known properties of speed, wavelength and amplitude. Furthermore, they reflect off a special particle, referred to as a wave center, that creates standing waves.

A fundamental wave traveling at a given speed and with a standard wavelength is what creates the universal clock at which everything can be measured against. When in motion, a change in an object's wavelength is the basis for relativity.

A fundamental wave amplitude creates the foundation for particle charge. Charge is the determining factor for the electrical force of particles and the movement, either attractive or repulsive, as particles respond to minimize wave amplitude. It is the principle cause of motion and forces. When particles are combined, their charges are added or subtracted due to constructive wave interference.

A fundamental wave center is the foundation for the creation of particles. As longitudinal waves reflect off the wave center to create standing waves, the stored energy becomes the mass of a particle. When wave centers combine, new particles are created. The stored energy in this new combination increases significantly, at the fifth power of the number of wave centers in combination, because wave amplitude and particle radius increase spherically in three-dimensional space.

The universe is simple and fundamental.

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